CINEAI LAB

Experimental Foundations for Open AI-Native Cinema

From Generated Clips to Image-Directed Filmmaking

DIRECTOR, CineLingo and an image-first architecture for controllable AI cinema

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https://cineai-lab.org/

Independent project. No affiliation with the tool or model providers discussed in this manuscript.

Status and reading note

This is a working research manuscript, not a product specification, a technical benchmark or a claim that a complete AI-native feature-film pipeline has been solved. CineAI Lab uses narrative projects as practical stress tests. The purpose is to document what current tools make possible, where they fail under filmmaking constraints, and what kind of authoring environment appears necessary when a creator wants continuity, selective revision and durable project memory.

> **The central change in version 0.10**
>
> The project has moved from asking how to generate convincing clips toward asking how to preserve a filmmaker's decisions while AI creates only the movement or transformation that has been authorised. This shift leads to an image-first model: persistent source images, shot instances, named selections, temporal behaviours and replaceable AI engines.

What changed since version 0.9

- The experimental record now includes local LTX-2.5 work and later ComfyUI / Wan 2.2 image-to-video tests.
- The working method has shifted toward validated still images as persistent project assets rather than disposable prompts.
- DIRECTOR is now described as a database-backed authoring interface under development, not merely as an abstract open-film aspiration.
- CineLingo is introduced as a natural-language orchestration layer inspired by the separation of Cast, Stage, Score and Behavior in Director/Lingo.
- The Missing Piece has become the principal image-first film demonstrator.
- The technology watch now focuses on first/last-frame control, interpolation, faster iteration, provenance and the integration of AI into creative workspaces.

Authorship and AI assistance

The protected narrative works, their characters, dramatic choices and final creative decisions remain the responsibility of their human author. ChatGPT has been used as a sustained research and development collaborator for screenplay analysis, experimental design, documentation, tool comparison, debugging, architecture work and editorial consolidation. This assistance does not transfer authorship of the narrative works and does not imply affiliation between CineAI Lab and OpenAI.

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Executive summary

Artificial intelligence can now generate striking still images, synthetic voices and short moving sequences. Yet the production of an editable, coherent film remains a different problem. A filmmaker does not only need a clip that looks plausible. The filmmaker needs to decide what the shot is, preserve characters and objects across time, direct performance, return to an earlier shot without losing continuity, and change one element without accidentally changing everything else.

CineAI Lab began by experimenting with the available components of this emerging ecosystem: still-image development, 2.5D camera motion, Python and FFmpeg pipelines, synthetic voice, multilingual visemes, facial-animation routes, local video generation with LTX, and later ComfyUI / Wan image-to-video workflows. Several experiments produced valid files. Fewer produced convincing cinema. This distinction between technical success and cinematic success became the research object.

The most important conclusion is architectural. In many current generative workflows, the model is treated as the place where the scene is created. The user provides an image or prompt and the model invents the next frames. For a filmmaker who has already designed the image, this can be backwards. If the face, hand position, composition, lighting, background and camera have already been chosen, regenerating the entire frame to create a small movement is both visually risky and computationally expensive.

CineAI Lab therefore proposes an image-first workflow. A film can be understood as a sequence of deliberately chosen still states, much like a highly sophisticated flipbook. The filmmaker may know the state at the beginning of a shot and the state at the end. The useful role of AI is then to create the missing movement between those states while respecting what has not changed.

> **Working proposition**
>
> Image A → selected movement → Image B. The filmmaker defines the images, the elements allowed to move, the elements that must remain locked and the intended timing. The AI engine executes the transition inside those limits.

This proposition led to the development of DIRECTOR, an interface concept that stores images as persistent assets, places them as instances in shots, attaches masks and locks to selected regions, represents movement as timeline events, and delegates execution to replaceable engines. CineLingo complements DIRECTOR as a human-readable orchestration layer. Rather than asking a creator to manipulate a node graph, it translates directions such as “at 12 seconds, move the right hand slowly; keep the camera fixed” into structured temporal actions and constraints.

The approach deliberately separates film, interface and engine. The film is the durable creative object: scenes, images, dialogue, timing, continuity, selections and behaviours. DIRECTOR is the authoring interface. LTX, ComfyUI, Wan, MuseTalk, Blender, FFmpeg or future models are execution engines and adapters. If one engine disappears or improves, the film should survive.

The Missing Piece is now the main practical demonstrator for this approach. It contains recurring characters across age and geography, family group scenes, intimate micro-performance, technical environments and a recurring puzzle motif. These elements make it useful for testing exactly the problems that generative video still struggles with: identity preservation, ensemble continuity, selective motion and the ability to revise a shot without redrawing the world.

Version 0.10 therefore reframes CineAI Lab. The project is no longer primarily searching for a single model capable of “making the film.” It is investigating a filmmaker-controlled environment in which multiple AI systems can be used as instruments while the project memory, visual decisions and dramatic structure remain under human control.

1. Cinema needs more than generated clips

1.1 A clip is an output; a film is a revisable system

Generative video demonstrations are often evaluated one clip at a time. Cinema, however, is cumulative. The credibility of shot 27 depends on what the audience saw in shots 2, 9 and 18. A costume detail, a piece of furniture, the age of a character, the direction of a gaze or the position of an object may carry narrative meaning. These elements must persist even when the tool used to render the shot changes.

Long-form work also demands revision. A director may want to change only the hesitation before a line, extend a hand gesture by half a second, hold the camera still, or replace one background object. A system that responds by regenerating the entire frame behaves less like an editing environment and more like a lottery whose ticket is a prompt.

1.2 The filmmaker already knows more than the model

In many practical situations the filmmaker is not starting from nothing. The character has already been designed. The composition has been approved. The lighting, lens feeling, decor and emotional beat have been chosen. The challenge is not invention but controlled transformation. This is why image-to-video is interesting but insufficient when the only strong anchor is the first frame.

The key filmmaking question is therefore not merely “Can the model generate motion?” but “Can the system preserve all decisions that are outside the requested motion?”

1.3 Human direction as explicit data

Human control is often described in broad ethical terms. CineAI Lab treats it as a concrete data problem. Direction should be represented explicitly: which character is present, which image is the approved source, which region may move, which region is locked, what happens at which time, what continuity rule must hold, and which kinds of failure should invalidate a render. If those decisions exist only inside a prompt, they are difficult to audit, revise or move between engines.

> **Design principle**
>
> The project should remember the director's decisions independently of the model that happened to render a particular version.

2. Experimental method: film problems first

2.1 Narrative projects as stress tests

CineAI Lab does not begin with a leaderboard of models. It begins with scenes. A scene imposes requirements that are difficult to hide: the same face must return, a hand must touch a specific object, a pause must last long enough, an animal must appear to have weight, or a group must react without the background melting. The narrative project supplies the pressure; the experiment reveals the technical gap.

2.2 The difference between file success and cinematic success

A render can complete without errors and still fail as cinema. Conversely, a modest technical result can be useful if it reveals a missing control. CineAI Lab therefore records both layers: whether the pipeline ran, and whether the result respected the intended shot.

Period	Experiment	Question	Main observation
Jun–Jul 2026	Still images and visual anchors	Can character and composition be fixed before motion?	Yes. Approved stills offered the strongest continuity and became reliable shot anchors.

Jul–Aug 2026	Blender / Python / FFmpeg 2.5D	Can controlled camera movement create cinematic life?	Camera and parallax were controllable, but motion of the frame did not equal actor performance.
Jul–Aug 2026	Voice, visemes, lip-sync	Can speech become directed performance?	Timing and mouth-shape data were feasible; gaze, breathing, anatomy and emotional rhythm remained harder.
Aug 2026	LTX-2.5 local inference	Can an open video model run locally on Apple Silicon?	Local inference was technically possible with aggressive offloading, but production control and speed remained limiting.
Sep–Oct 2026	ComfyUI / Wan 2.2 I2V	Can a validated image become a stable moving shot?	The tested workflow used a starting image plus prompt; whole-shot reinterpretation and long iteration times undermined precise direction.
Oct 2026	DIRECTOR / CineLingo	What should the authoring layer remember?	Images, instances, selections, locks, timing and behaviours should persist independently of the rendering engine.

2.3 Hardware realism matters

Experiments were deliberately performed on consumer-class hardware as well as through hosted tools. This is important because a filmmaker-oriented workflow cannot be judged only on an ideal high-memory GPU. Waiting many minutes for a few seconds of uncertain output changes the creative process itself. Direction depends on iteration, and iteration depends on feedback speed.

3. What the experiments taught us

3.1 Still images: the most stable source of authorship

The still image repeatedly proved to be the point at which the filmmaker had the clearest control over composition, character identity and production design. Once a still was approved, treating it as disposable made little sense. The natural conclusion was to elevate approved images from temporary prompts to persistent project assets.

3.2 2.5D motion: precise but not performance

Local 2.5D tests confirmed that camera motion, parallax, compositing and export can be made deterministic. This is valuable for transitions, atmospheric movement and certain documentary or graphic styles. But it also exposed a conceptual limit: moving a camera over a still image does not make a character act. The body needs internal movement, weight transfer, gaze and timing.

3.3 Voice and facial animation: timing is easier than acting

Synthetic speech, viseme sequences and lip-sync pipelines can be structured and timed. The difficult part is not the existence of mouth movement but the coherence of performance. Speech includes breath, hesitation, listening, gaze, posture and emotional intention. These dimensions need to be directed together rather than generated as unrelated effects.

3.4 LTX-2.5: local inference was possible, but not yet a complete production path

The August 2026 LTX-2.5 experiment demonstrated that local inference on a 24 GB Apple M3 could complete through a native Python pipeline with aggressive offloading, even though the combined model components exceeded available unified memory. This was an important technical result. It did not, however, solve the filmmaking problem: generation remained slower than interactive direction and the resulting motion did not consistently preserve the desired image with sufficient precision for the current film.

> **Interpretation**
>
> A successful inference proves that a model can run. It does not prove that it belongs at the centre of a filmmaking interface.

3.5 ComfyUI / Wan 2.2: the cost of regenerating too much

Later tests moved to ComfyUI and Wan 2.2 image-to-video workflows. In the tested configuration, the workflow accepted a starting image and a textual description of what should happen next, but it did not provide a true filmmaker-defined final image as a second hard anchor. The model therefore had to invent the continuation.

For the kinds of shots being developed in The Missing Piece, this was often the wrong trade-off. A desired micro-action could be accompanied by face drift, altered hands, changed objects, new camera motion or a reinterpretation of the background. The problem was amplified by generation time: a few seconds of footage could require enough setup and rendering time that repeated small corrections became impractical.

The conclusion is deliberately limited to the workflows actually tested. It is not a claim that ComfyUI or Wan cannot support other forms of control. It is a record of why that path was not accepted as the central CineAI Lab production method at this stage.

3.6 The useful failure

These experiments did not merely eliminate tools. They clarified the missing layer. The core requirement is not another generator. It is an authoring environment capable of telling any generator what it is allowed to change, what it must preserve, and where the shot must arrive.

4. The image-first turn: film as a flipbook

4.1 The simplest mental model

At a fundamental level, moving images are sequences of still frames. The analogy is intentionally simple: a film is a very sophisticated flipbook. Traditional cinema captures or constructs those frames through photography, animation, visual effects and editing. Generative AI adds a new way to create intermediate frames, but it does not remove the need for endpoints, continuity and directorial choice.

```
IMAGE A  →  SELECTED MOVEMENT  →  IMAGE B
```

This model changes the role of AI. Instead of treating the first image as a suggestion and the remainder of the shot as free space, both images can be treated as deliberate states. The engine's job becomes to construct a plausible, temporally coherent path between them.

4.2 Movement should be local when the intention is local

A father may point at a missing puzzle piece while the table, background and other family members remain largely unchanged. A child may shift his gaze while the camera stays fixed. A hand may return to a lectern after a sentence. These are local events. Recomputing every pixel as if the entire shot were uncertain is wasteful and invites drift.

4.3 Start image, end image and intermediate keyframes

The preferred workflow therefore includes at least a start image and an end image, with optional intermediate keyframes when the action is complex. The keyframes are not simply conditioning images; they are directorial commitments. The system should treat them as states to be respected.

4.4 Selected motion and locked regions

The filmmaker should be able to select a region or object and declare that it may move, while simultaneously locking identity, costume, framing, camera and background. This transforms control from a vague prompt into an explicit contract.

> **Image-first proposition**
>
> The filmmaker assembles the important states first. AI supplies continuity between them, not permission to redesign the shot.

5. DIRECTOR: a filmmaker-facing authoring environment

5.1 Why the name matters

DIRECTOR is not used here as a generic synonym for “control.” It refers to an interface architecture being developed within CineAI Lab, inspired by an earlier authoring insight found in Macromedia/Adobe Director: separate the source media, the visible composition, the temporal score and the behaviours that act on instances.

5.2 From Director/Lingo to CineAI

Director / Lingo	CineAI DIRECTOR	Function
Cast	Media / Assets	Persistent source images, video, audio, characters, sets and props
Stage	WYSIWYG Stage	Exact visible composition of the shot
Cast member	Source media	Original asset preserved independently of placement
Sprite	Shot instance	Placed use of an asset with its own transform, timing, mask and locks
Score	Sequence / Timeline	Temporal organisation of events and instances
Behavior	Cinematic behaviour	Reusable performance or motion instruction
Lingo	CineLingo	Human-readable orchestration translated to engine-specific actions
Playback head	Playhead	Current temporal position inside the shot
Movie	Project / film	Durable creative object independent of rendering engine

5.3 Four visible zones

A practical DIRECTOR interface can be organised around four visible zones without copying Director's historical appearance:

- Media / Cast: the available images, audio, characters, sets and props.
- Stage: the exact WYSIWYG composition, with selectable and movable instances.
- Inspector: masks, feathering, locks, animatable regions and continuity metadata.
- Score / Timeline: dialogue cues, gestures, gaze, expression, camera, audio, effects and lock events.

5.4 The Stage is the visual truth

The decisive rule is that the Stage represents the filmmaker's intended image. The AI may animate selected properties, but it should not silently redefine the composition. This borrows the useful distinction from Director between what the author lays out visually and what a behaviour may modify at runtime.

5.5 AI belongs behind the interface

```
Filmmaker
   ↓
DIRECTOR Stage + Timeline + Constraints
   ↓
Structured CineAI representation
   ↓
Engine adapter
   ↓
LTX / Wan / ComfyUI / MuseTalk / Blender / future model
   ↓
Render
```

The creative object should therefore remain readable even if a specific engine disappears. DIRECTOR should not become a simplified copy of ComfyUI. ComfyUI, when useful, should sit behind DIRECTOR as one execution route among others.

6. A database model for film memory

6.1 Why a film needs structured memory

Long-form continuity cannot depend on a folder of prompts. The project needs a durable representation of source media, their uses in individual shots, the events that occur over time, the constraints applied to those events, and the renders produced by different engines.

6.2 Proposed core entities

Entity	Purpose
assets	Persistent source media: images, video, audio, masks, references
shots	The dramatic and temporal unit being directed
shot_instances	A placed use of an asset inside a shot
dialogues	Text actually spoken, language, speaker and timing
instructions	Positive directorial instructions
negative_constraints	Errors or transformations that must be avoided
timeline_events	Timed behaviours such as gesture, gaze, expression, camera or lock
renders	Generated or composited outputs and their lineage
engine_jobs	Execution records for a specific rendering engine and configuration

6.3 Source asset versus shot instance

The distinction between source and instance is essential. A source image should not be mutated simply because it is placed in a shot. The instance owns its own position, size, rotation, opacity, timing, mask, feathering, locks and animatable regions.

```
shot_instance
  shot_id
  asset_id
  instance_name
  x_pct / y_pct
  width_pct / height_pct
  rotation_deg / z_index / opacity
  start_ms / end_ms
  mask_uri / feather_px
  locked_identity / locked_costume / locked_position / locked_background
  animatable_regions
  metadata
```

6.4 Timeline events as first-class data

```
timeline_event
  shot_id
  instance_id
  event_order
  event_type
  label
  start_ms / end_ms
  instruction
```

```
    parameters
    status
```

Typical event types include gesture, gaze, expression, head motion, camera, audio, music, lock, unlock, mask, behavior, dialogue cue, hold, transition and visual effect. This representation makes a director's intentions queryable and portable.

6.5 Why a database is different from a node graph

A node graph describes how one execution is assembled. A film database describes what the film is supposed to be. The two are related but should not be identical. Keeping them separate prevents the project from becoming trapped inside a specific workflow implementation.

> **Durability rule**
>
> A render can be disposable. A directorial decision should not be.

7. CineLingo: behaviours instead of prompt piles

7.1 A language for cinematic events

CineLingo is conceived as a filmmaking orchestration layer, not as a general-purpose programming language. The filmmaker describes cinematic events in natural language. CineLingo resolves the affected instances, temporal references, locks and animatable regions, then delegates execution to the appropriate adapter.

```
Natural language
      ↓
CineLingo parser
      ↓
CineLingo core
      ↓
Stage + Timeline + Constraints
      ↓
Engine adapters
      ↓
Render
```

7.2 Three levels of expression

The system distinguishes the human direction, the structured CineLingo representation, and the engine-specific implementation. The filmmaker should normally operate only at the first level.

Level	Example
Director	“At the end, he looks at the camera with concern.”
CineLingo core	BEFORE dialogue.end 1.5s → gaze CAMERA → expression concerned 0.35
Engine adapter	Model-specific masks, prompts, controls, samplers, keyframes or compositing instructions

7.3 Locks and animatable regions

```
LOCK { identity costume background framing camera }

ANIMATE ACTOR { face head arms hands }
```

This is not cosmetic syntax. It expresses the key contract between director and model. A request for a hand gesture should not imply permission to change the face, costume or camera.

7.4 Temporal direction

```
AT 12s
  ACTOR.hand.right -> OPEN
  DURATION 2s
  EASE gentle
END

AT 15s
  ACTOR.gaze -> CAMERA
  DURATION 1.2s
END

FROM 18s TO END
  LOCK CAMERA
END
```

7.5 Contradiction detection before generation

A structured language also permits validation. If the camera is locked and a later instruction requests a pan, the interface can flag the contradiction before expensive generation begins. The same is true for a frozen hand followed by a gesture request. Preventing impossible or contradictory jobs is one way to reduce wasted computation.

7.6 Behaviours as reusable direction

CineLingo behaviours can encode recurring acting patterns without freezing them into a single prompt. A “restrained speaker” behaviour might imply subtle head motion, low hand activity, serious expression and audience-facing gaze. Each engine may implement those intentions differently, while the behaviour remains part of the film project.

8. Film, interface and engine must remain separate

8.1 Three layers

Layer	What it contains	What should survive
Film	Scenes, shots, validated images, dialogue, timing, continuity, selections, behaviours	Everything creatively meaningful
Interface	DIRECTOR Stage, Inspector, Score, editing tools	The authoring logic and project readability
Engine	LTX, Wan, ComfyUI workflows, MuseTalk, Blender, FFmpeg, future systems	Replaceable implementation details and render parameters

8.2 Why engine independence matters

The AI video field changes quickly. Models appear, are replaced, change licensing, move between hosted and local forms, or disappear. A feature-length project may outlive several generations of tools. Treating the engine as a replaceable backend is therefore not only an open-source preference; it is a basic production-resilience requirement.

8.3 Generation should be a job, not the project

An engine job should record which shot version, assets, masks, constraints, behaviour events and engine parameters were used to produce a render. The job can fail, be retried or be superseded without damaging the creative source of truth.

8.4 Selective re-rendering

This architecture also makes selective revision possible. If a gesture changes, DIRECTOR should be able to identify the affected region and time interval, create a new engine job, and preserve the rest of the project. The long-term goal is not necessarily pixel-level partial diffusion in every engine; it is a workflow in which the scope of the requested change remains explicit.

9. The Missing Piece as the current demonstrator

9.1 Why this film is useful technically

The Missing Piece is the current primary film project for testing image-first production. Its public description is intentionally limited because the narrative work remains protected. Technically, however, the project is demanding in productive ways: the same protagonist appears across time and geography; family and research-group scenes require ensemble continuity; a recurring puzzle image carries visual meaning; and later sequences move into high-technology environments where small mechanical or biological changes may matter.

9.2 The opening puzzle sequence

The opening sequence provides a compact demonstration of the DIRECTOR problem. A family is gathered around a puzzle. A father notices the missing place. Attention shifts. The family leaves for a piñata. The child Mateo later finds a piece and returns to the puzzle. Much of the shot design already exists as validated images.

For a transition between two near-identical family images, the desired change may be only a hand movement, a pointing gesture and subtle changes of gaze and expression. The table, room, camera and most bodies should remain stable. This is precisely the kind of shot where whole-frame regeneration is an inefficient abstraction.

> **Demonstrator question**
>
> Can a system create the missing frames between two deliberately designed images while changing only the selected actions and preserving the rest of the shot?

9.3 Continuity over years and locations

Later scenes follow Mateo through different periods of life and across Guatemala and Sweden. The challenge is not to force identical pixels across decades, but to preserve identity cues and narrative continuity while allowing age, costume and context to change deliberately. The database therefore needs both invariants and versioned character states.

9.4 Technical environments and controlled complexity

The project also includes scientific and technological environments, including a biological computing concept. These scenes test whether the system can preserve designed machinery and spatial relationships while directing local reactions or state changes. Again, the requirement is selective transformation, not automatic redesign.

9.5 A production consequence

The immediate production strategy is therefore pragmatic. The film can first exist as a carefully timed sequence of validated images, dialogue, sound, cuts and selected camera effects. Motion is added only where it contributes. This creates a coherent film structure before asking generative systems to animate every frame.

10. Openness, rights, provenance and human direction

10.1 Open does not mean model-free

An open filmmaking environment may use proprietary or hosted models when useful. Openness here refers primarily to the durability and intelligibility of the project: standard media files, exportable metadata, explicit rights and provenance, replaceable adapters, and the ability to continue editing if a provider disappears.

10.2 Provenance should be part of production, not an afterthought

As synthetic media becomes normal, a film project should be able to record which assets were human-created, AI-assisted, generated, licensed, transformed or composited; which model or service produced a render; and which human decisions governed the output. Provenance is therefore closely related to the same database architecture needed for continuity.

10.3 Human direction should be inspectable

A human-in-the-loop claim is weak if the human contribution exists only as a vague prompt. DIRECTOR and CineLingo aim to make direction inspectable: selected images, explicit masks, locks, timing, behaviours, rejected renders and approved states. This is both a creative and governance advantage.

10.4 European context

European transparency requirements for certain AI-generated or manipulated content became applicable in August 2026. The exact legal treatment depends on context and implementation, but the broader production lesson is already clear: provenance, disclosure and documentation are becoming part of audiovisual workflow design rather than optional metadata added at the end.

11. Technology watch: October 2026

The most relevant signals for CineAI Lab are not simply improvements in spectacle. They are signs that the ecosystem is moving toward stronger anchoring, faster iteration, integrated editing and engine abstraction.

Date	Signal	Why it matters for CineAI Lab
13 Oct 2026	ComfyUI / NVIDIA VFX frame-generation nodes interpolate between adjacent frames.	Existing frames are treated as anchors, which is conceptually closer to the flipbook model than unconstrained whole-shot invention.
2 Oct 2026	Human Provenance in Film consultation update tests production declarations.	Provenance can be designed into a project database rather than handled only at distribution.
30 Sep 2026	Comfy API packages workflows as deployable services.	The execution layer can move behind a filmmaker-facing interface.
28 Sep 2026	Wan 2.7 documentation lists first-frame, first-and-last-frame, and continuation modes.	True first/last-frame anchoring moves closer to the desired keyframe workflow.
23 Sep 2026	Comfy Router exposes multiple media models through one API.	Model abstraction supports engine independence.
10 Sep 2026	Runway research targets lower time-to-first-frame and streaming generation.	Fast feedback is essential for repeated small directorial corrections.

8 Sep 2026	Adobe Firefly beta integrates generation, element-level editing, assets and video assembly.	AI is moving from isolated generators toward structured creative workspaces.
12 Aug 2026	LTX-2.5 expands open local video workflows.	Local execution remains strategically important even when current quality or speed is insufficient.
31 Jul 2026	Seedance 2.5 emphasises reference-based multimodal generation and longer sequences.	Reference control and sequence-level thinking are becoming mainstream model objectives.
2 Aug 2026	EU AI transparency obligations become applicable.	Synthetic-media provenance increasingly intersects with production architecture.

11.1 A field moving toward the missing layer

These developments do not prove that the DIRECTOR problem is solved. They do suggest convergence. First/last-frame control, interpolation, streaming generation, unified workspaces and multi-model APIs all reduce the distance between today's generators and a true filmmaking environment. The remaining question is who owns the project memory and directorial logic: the model interface, or the filmmaker.

11.2 Selected current sources

ComfyUI, NVIDIA VFX nodes: https://blog.comfy.org/p/nvidia-vfx-nodes-for-video-upscaling

Human Provenance in Film: https://humanprovenance.film/news/consultation-update-september/

ComfyUI blog archive / API announcements: https://blog.comfy.org/archive

Alibaba Cloud, Wan image-to-video guide: https://www.alibabacloud.com/help/en/model-studio/wan-image-to-video-guide

Comfy Router: https://blog.comfy.org/p/introducing-comfy-router-one-api

Runway, Towards instant video generation: https://runway.com/news/research/towards-instant-video-generation

Adobe Firefly unified generation and editing: https://helpx.adobe.com/fr/firefly/web/unified-generation-and-editing-experience/generation-and-editing-experience-overview.html

ComfyUI / LTX-2.5 support: https://blog.comfy.org/p/ltx-25-day-0-support-in-comfyui

ByteDance Seedance 2.5: https://seed.bytedance.com/en/blog/one-take-creation-flexible-referencing-introducing-seedance-2-5

European Commission, AI Act transparency guidelines: https://digital-strategy.ec.europa.eu/en/library/guidelines-transparency-obligations-providers-and-deployers-ai-systems

12. Evaluation framework for controllable AI filmmaking

12.1 Quality should be measured against directorial intent

A beautiful output can still be a failure if it changes the wrong thing. Evaluation should therefore compare the render against explicit shot constraints, not only against generic visual quality.

Dimension	Question	Example acceptance criterion
Identity	Did the intended character remain recognisable and consistent?	No unintended face drift; age changes only when specified.
Composition	Did the camera and spatial arrangement remain as directed?	Locked camera and framing remain unchanged.
Object permanence	Did props and designed objects survive?	Puzzle piece, table or machinery remains present and stable unless intentionally moved.
Selective motion	Did only authorised regions move?	Hand gesture occurs without unrelated background or costume change.

Temporal accuracy	Did actions occur at the intended time and duration?	Gaze begins within the specified cue window.
Performance	Does motion read as intentional acting rather than noise?	Pause, gaze and gesture support the dramatic beat.
Continuity	Does the shot remain compatible with adjacent shots?	Costume, lighting and object state match the sequence.
Iteration cost	Can the filmmaker revise the shot at practical speed?	Small changes do not require rebuilding unrelated content.
Provenance	Can the origin and transformation history of assets be reconstructed?	Render links back to source assets, instructions and engine job.
Portability	Can the project survive a change of engine?	Core shot representation does not depend on one model graph.

12.2 Failure should be informative

A useful production system should not only say that a render failed. It should identify what constraint was violated: identity drift, camera movement, hand anatomy, background deformation, temporal mismatch or another category. This transforms rejected generations into structured evidence for both creative iteration and model evaluation.

13. Roadmap

13.1 Near-term: finish the film structure before perfecting animation

- Consolidate The Missing Piece as a complete image-first sequence with validated images, durations, dialogue and sound.
- Keep the canonical one-image-per-shot/idea discipline and 16:9 framing for film assets.
- Use dedicated close-up images rather than cropping or reusing wide shots when the dramatic function changes.
- Animate selectively, beginning with shots where movement clearly adds information or emotion.

13.2 DIRECTOR prototype

- Complete the persistent asset and shot-instance model.
- Keep selections named, movable, removable and stored independently from source images.
- Expose masks, feathering, locks and animatable regions in the Inspector.
- Represent dialogue, instructions, errors to avoid and timeline events as separate editable layers.
- Allow engine adapters to consume the same shot representation.

13.3 CineLingo development

- Parse natural-language direction into explicit objects, time references and actions.
- Validate contradictions before rendering.
- Maintain a reusable behaviour library for common cinematic performance patterns.
- Compile the same behaviour into different engine-specific implementations.
- Preserve the filmmaker-facing description even when an engine changes.

13.4 Engine research

- Test true first/last-frame workflows as they become practically accessible.
- Compare interpolation approaches with generative transition approaches for local micro-movement.
- Measure time-to-first-preview and revision latency, not only final visual quality.
- Prefer engines that expose region control, keyframes, camera locks and deterministic constraints.
- Avoid rebuilding the CineAI architecture around a single rapidly changing model family.

13.5 Research questions

- How much of a shot can remain literally unchanged while selected regions are generated?
- Can first/last-frame constraints preserve identity better than start-image-only conditioning?
- What representation of gaze, gesture and expression is portable across video models?
- How should an authoring tool distinguish hard locks from soft preferences?
- Can partial re-rendering reduce both generation cost and continuity errors?
- Which provenance data should be stored by default for an AI-assisted film?
- What is the minimum interface a filmmaker needs to direct AI without becoming a node-graph engineer?

Conclusion

CineAI Lab began with the excitement and frustration of emerging AI media tools. The experiments confirmed that images, speech and motion can increasingly be generated. They also made a different problem impossible to ignore: filmmaking depends on continuity of intention.

The central proposal of version 0.10 is therefore not a new generative model. It is an authoring architecture. Approved images become persistent assets. Their uses become shot instances. Selections, masks and locks define what may change. CineLingo describes events over time. AI engines execute behind the interface and can be replaced. The film survives them.

This is a modest claim compared with promises of automatic cinema, but it may be a more durable one. The goal is not to ask artificial intelligence to become the director. The goal is to give the director a better instrument.

> **CineAI Lab, October 2026**
>
> The film is the project. The interface remembers the direction. The engine performs a task.

Appendix A. Director/Lingo to CineAI mapping

Historical concept	CineAI interpretation	Reason retained
Cast	Media / Assets	Separates source resources from their uses in shots.
Stage	WYSIWYG Stage	Makes the intended composition explicit and editable.
Score	Timeline / Sequence	Represents when instances and events exist.
Cast member	Source media	Preserves originals independently from shot placement.
Sprite	Shot instance	Gives each placed use its own transform, timing and constraints.
Behavior	Cinematic behaviour	Encodes reusable acting or motion intentions.
Frame script	Temporal event	Attaches actions to moments or intervals.
Sprite script	Instance behaviour	Targets one placed character or object.
Movie script	Project behaviour	Represents project-wide rules or policies.
Lingo	CineLingo	Adds structured behaviour when visual layout alone is insufficient.
Playback head	Playhead	Keeps time as a native part of direction.
Property list	Structured parameters	Supports engine-neutral representation of constraints.

Appendix B. Minimal data model

```
assets(id, project_id, type, uri, version, provenance, rights, metadata)

shots(id, scene_id, code, title, duration_ms, status)

shot_instances(id, shot_id, asset_id, instance_name, x_pct, y_pct, width_pct, height_pct,
rotation_deg, z_index, opacity, start_ms, end_ms, mask_uri, feather_px, locks,
animatable_regions, metadata)

dialogues(id, shot_id, speaker, language, text, start_ms, end_ms)

instructions(id, shot_id, scope, text, priority)

negative_constraints(id, shot_id, scope, text, severity)

timeline_events(id, shot_id, instance_id, event_order, event_type, label, start_ms,
end_ms, instruction, parameters, status)

renders(id, shot_id, engine_job_id, uri, approved, quality_notes, provenance)

engine_jobs(id, shot_id, adapter, engine, model, config, started_at, finished_at, status,
logs)
```

This schema is intentionally conceptual. Implementation details may evolve, but the separation between durable film data and replaceable engine execution should remain.

Appendix C. Experiment matrix

Experiment	What worked	Observed limit	Architectural lesson

Validated stills	Strong control of image identity, composition and production design	No inherent motion	Treat approved images as persistent assets.
2.5D camera / parallax	Deterministic camera and compositing	Read as a moving still when performance was needed	Separate camera motion from actor/object behaviour.
Synthetic voice	Speech generation and timing	Performance nuance remained generic in difficult scenes	Dialogue and acting direction need separate layers.
Visemes / lip-sync	Mouth-shape timing can be structured	Facial anatomy and identity instability	Use region-specific control and performance constraints.
LTX-2.5 local	Open local inference technically possible	Memory pressure, blur/camera instability, iteration speed	Local execution matters, but engine must remain replaceable.
ComfyUI / Wan 2.2 tested workflow	Real I2V clips from a source image	Start-image-only control, whole-shot drift, long iterations	Need end-image anchors, locks and selected motion.
DIRECTOR / CineLingo	Film can be represented as assets + instances + behaviours	Interface and adapters still under development	Move the creative source of truth above the engine layer.

Appendix D. Selected references

CineAI Lab public site: https://cineai-lab.org/

CineAI Lab GitHub repository: https://github.com/ReneBoroughs/cineai-lab

Adobe Director Support Center archive: https://www.adobe.com/support/director/downloads.html

Director Scripting Reference, Macromedia Director MX 2004: https://eclass.hmu.gr/modules/document/file.php/TP194/drmx2004_scripting_ref.pdf

Cardiff University, Lingo / Director basics: https://users.cs.cf.ac.uk/Dave.Marshall/Multimedia/PDF/Lingo_Basics.pdf

ComfyUI blog: https://blog.comfy.org/

Alibaba Cloud Wan image-to-video documentation: https://www.alibabacloud.com/help/en/model-studio/wan-image-to-video-guide

Runway Research: https://runway.com/research

Adobe Firefly: https://www.adobe.com/products/firefly.html

European Commission AI Act transparency guidance: https://digital-strategy.ec.europa.eu/en/library/guidelines-transparency-obligations-providers-and-deployers-ai-systems

Human Provenance in Film: https://humanprovenance.film/

Project documents informing this edition

- CINEAI_DIRECTOR_LINGO_WYSIWYG_REFERENCE.md — working architecture reference, 16 August 2026.
- CINELINGO_SPEC_V01.md — initial functional specification, 1 September 2026.
- CineAI_Lab_LTX25_Apple_Silicon_Technical_Report.docx — local inference experiment, 15 August 2026.
- The Missing Piece storyboards and scene documents — October 2026 working materials.
- CineAI Lab website research, experiment and watch pages — updated 18 October 2026.

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